Data Governance for the UCI Adult Dataset

Assignment No. 5

1. Purpose of data governance for this dataset

The Adult dataset (48,842 census records) is used to study income. Its records describe people and include sensitive attributes: race, sex, and capital gains and losses. Governance sets the rules that make the data fit for use and protects the people in it. For this dataset that means four things:

- **Trust in results.** Income studies are only as good as the data. Label inconsistencies and hidden missing values change the answers, so they must be found and fixed in a documented, repeatable way.

- **Privacy.** Demographic combinations can single out individuals. Quasi-identifiers must be generalised and small groups suppressed before the data is shared.

- **Accountability.** Every value that changes must be traceable: source checksum, rules applied, rows affected.

- **Operational control.** A scheduled, monitored pipeline that stops on failure, so governance does not depend on someone running scripts by hand.

2. Metadata management

Source: https://archive.ics.uci.edu/dataset/2/adult. Source file SHA-256: `26b38f8623766396e2ba0d1a235a283db8ed993cc32bf32fc8aab09708c3bdf2`. Rows: 48,842 raw, 48,790 clean, 45,165 published. Classification of the 15 raw columns: quasi-identifier 9, non-sensitive 1, sensitive attribute 4, quasi-identifier and sensitive attribute 1. Owner and retention are placeholders in `dataset_metadata.json` until a data governance council assigns them.

Column	Type	Description	Sensitivity	Nullable	Published as
age	integer	Age in years	quasi-identifier	no	generalised to 10-year bands (age_band)
workclass	categorical	Type of employer	quasi-identifier	yes	kept; missing values set to Unknown
fnlwgt	integer	Census sampling weight (describes the sample, not a person)	non-sensitive	no	removed: not needed for the governance use case
education	categorical	Highest education level	quasi-identifier	no	removed: redundant with education-num
education-num	integer	Years of education, ordinal 1-16	quasi-identifier	no	kept
marital-status	categorical	Marital status	quasi-identifier	no	kept
occupation	categorical	Occupation	quasi-identifier	yes	kept; missing values set to Unknown
relationship	categorical	Family role in household	quasi-identifier	no	kept
race	categorical	Race (self-reported)	sensitive attribute (special category where GDPR applies)	no	kept; used only inside aggregate analysis
sex	categorical	Sex (Male or Female)	quasi-identifier and sensitive attribute	no	kept
capital-gain	integer	Capital gains in the year (USD)	sensitive attribute (financial)	no	kept; flag added where value is 99999 (top-code)
capital-loss	integer	Capital losses in the year (USD)	sensitive attribute (financial)	no	kept
hours-per-week	integer	Usual hours worked per week	quasi-identifier	no	kept
native-country	categorical	Native country	quasi-identifier	yes	generalised to United-States or Non-US (native_region)
income	categorical	Income above or below 50K USD per year	sensitive attribute	no	kept; trailing period removed

The dictionary is the metadata catalogue for this dataset. Apache Atlas could hold the same records for a larger estate; this project keeps them as CSV and JSON so they can be checked in version control.

3. Data quality assessment and improvement

Each rule counts rows that break it. The profile runs on the raw file, then on the cleaned table, using the same rules. Rules that find nothing still appear with zero, so the assessment shows what was checked.

Rule	Rows failing before	Rows failing after
missing_workclass	2,799	0
missing_occupation	2,809	0
missing_native_country	857	0
income_trailing_period	16,281	0
income_invalid_value	0	0
exact_duplicate_rows	29	0
whitespace_in_text_cells	0	0
age_out_of_range	0	0
hours_out_of_range	0	0
education_num_out_of_range	0	0
negative_capital_values	0	0
education_to_education_num_conflicts	0	0
capital_gain_topcode_99999	244	244



Improvements made, with the number of rows changed:

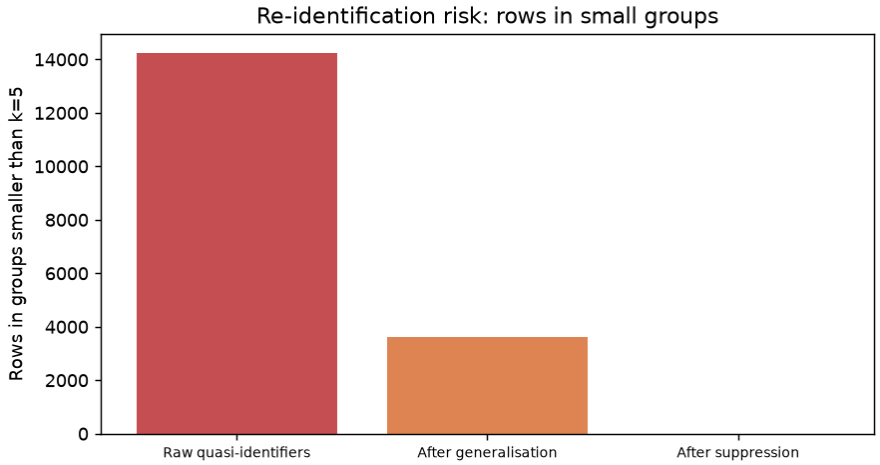
Rule	Column	Rows changed	Fix
missing_workclass	workclass	2,799	kept the row, set missing value to 'Unknown' (not dropped)
missing_occupation	occupation	2,809	kept the row, set missing value to 'Unknown' (not dropped)
missing_native_country	native-country	857	kept the row, set missing value to 'Unknown' (not dropped)
income_trailing_period	income	16,281	removed trailing period so both splits use the same labels
exact_duplicate_rows	all	52	removed exact duplicate rows (after labels were normalised)
capital_gain_topcode_99999	capital-gain	244	value kept; flag column capital_gain_topcoded added (possible top-code, not corrected)

Examples. (1) In the raw file, 2,799 workclass values are missing. Two markers are used: ? (1,836 rows) and the text `NaN` (963 rows). The first profile counted only the first. Both are now recognised, and the rows are kept with the value *Unknown*. (2) The test split labels income as `<=50K.` with a trailing period, 16,281 rows. 29 exact duplicates are in the raw file. Once labels and missing markers are normalised, more rows match, so 52 duplicate rows are removed in total. (3) 244 rows have capital gain 99999. That looks like a top-code, so the value is kept and a flag column is added rather than changing the number.

4. Privacy and security

Classification. The dataset has no direct identifiers (no names, IDs or addresses); the checks confirm none of the columns carries one. Quasi-identifiers (age, sex, race, marital status, native country, occupation, relationship, education, hours) could single someone out when combined. Income, race and capital gains and losses are sensitive attributes.

Re-identification risk. Quasi-identifiers were measured as a k-anonymity check: the smallest group of records sharing the same values. Before protection, the smallest group has 1 record(s), and 14,230 rows (29.2%) sit in groups smaller than 5. Age was generalised to 10-year bands and native country to United-States or Non-US. Groups still smaller than k are suppressed. The published file has k_min = 5 and removes 3,625 rows (7.4%).



Measure	Raw (cleaned)	Generalised only	Published
Smallest group size (k_min)	1	1	5
Groups	12,097	3,055	1,010
Rows in groups below k	14,230	3,625	0
Rows published	48,790	48,790	45,165

Data minimisation. education (duplicate of education-num), fnlwgt (a census sampling weight that describes the sample, not a person), and the raw age and native-country (replaced by the generalised columns) are not published.

Access and security design (documented; configured in the AWS definition where stated):

Control	Design	Status
Access roles	Analysts read only the published file. The governance steward alone reads the raw file (least privilege).	Designed. Not applied to any account.
Encryption at rest	S3 buckets with SSE-KMS default encryption. Keys managed by the data owner.	Designed. Bucket settings are outside the pipeline definition.
Encryption in transit	HTTPS only (bucket policy denies insecure transport).	Designed.
Public access	Block all public access on the governance buckets.	Designed.
Audit	Pipeline logs written to S3 (pipelineLogUri) and kept with the run summary.	In the AWS definition; logs are written locally here.
Failure alerting	SNS alarm on failed activity.	In the AWS definition; topic not created.

5. Orchestration

The pipeline runs as ordered steps. A failed step stops the run and is logged, which is how an AWS Data Pipeline activity behaves when it fails. Steps and durations from this run:

Step	Status	Seconds
ingest	succeeded	0.384
profile_before	succeeded	0.231
clean	succeeded	0.303
profile_after	succeeded	0.209
protect	succeeded	0.135
publish	succeeded	0.186

AWS Data Pipeline definition. `aws/datapipeline_definition.json` defines the same steps: an S3 input node (raw data), a ShellCommandActivity on a t2.micro EC2 resource that runs `pipeline.py`, an S3 output node (protected data), S3 pipeline logs, and an SNS failure alarm. It was not deployed, because the project has no AWS credits. It can be checked before deployment with `aws datapipeline validate-pipeline-definition`. AWS Data Pipeline is an older orchestration service; check that it is available to your account, and consider AWS Step Functions or Amazon MWAA for new work.

Screenshots. The deliverables ask for screenshots of the privacy and security measures. None are included, because no AWS console was used. The local outputs (logs, privacy metrics, quality tables) serve as the evidence instead.

6. Limitations

- The dataset is public and already anonymised by its providers. The privacy work shows the method, not a real re-identification risk assessment.

- Owner, retention period and the governance council are placeholders.

- The 99999 capital-gain value is flagged, not corrected. Whether it is a true top-code needs the data owner's confirmation.

- The AWS definition and access design were not deployed or tested in an account.

- The k threshold (5) and the age bands are policy choices, and they can be changed in governance.py.

7. References

- UCI Machine Learning Repository, Adult dataset: https://archive.ics.uci.edu/dataset/2/adult

- Sweeney, L. (2002). k-anonymity: a model for protecting privacy. International Journal on Uncertainty, Fuzziness and Knowledge-Based Systems, 10(5), 557-570.

- Apache Atlas, metadata and governance platform: https://atlas.apache.org

- AWS, Data Pipeline Developer Guide: https://docs.aws.amazon.com/datapipeline/latest/DeveloperGuide/what-is-datapipeline.html